

HEMANTH NAGESH

Software Engineer – Generative AI & Backend Systems

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PROFESSIONAL SUMMARY

Software Engineer with 2.5+ years of experience designing, building, and deploying Generative AI and LLM-powered backend systems in enterprise settings. Proficient in Python, LangChain, LangGraph, RAG architectures, prompt engineering, and vector databases — with hands-on experience integrating OpenAI APIs, Azure AI Foundry, and AWS Bedrock. Proven track record delivering measurable production outcomes, with additional research depth in accessible AI interfaces (WCAG, TCS Accessibility Team). Azure AI Engineer Certified.

TECHNICAL SKILLS

Programming: Python, Java, SQL

Generative AI & LLM: LangChain, LangGraph, AutoGen, OpenAI APIs, Prompt Engineering, LLM Evaluation Frameworks, RAG Architectures, Embedding Techniques, Semantic Search, Document Retrieval

AI Infrastructure & MLOps: vLLM (LLM Inferencing), Model Context Protocols, CI/CD for AI, LLMops, Responsible AI Practices, Performance & Cost Optimization

Backend & Systems: Django, REST APIs, Microservices, Distributed Systems, Containerization (Docker), Kubernetes, Fault Tolerance

Frontend: React.js, AI Interface Design Patterns, Accessible UI (WCAG Standards)

Databases & Storage: PostgreSQL, Vector Databases, Structured & Unstructured Data Processing

Cloud & DevOps: Azure AI Foundry, Azure OpenAI, AWS Bedrock, GCP Vertex AI, Docker, CI/CD, GitHub

EXPERIENCE

Tata Consultancy Services (TCS)

Software Engineer – Generative AI & Agentic Systems

Dec 2023 – Present

- Designed and deployed a production-grade multi-agent LLM backend using LangGraph and LangChain, reducing system execution time by 70% through parallel tool-calling, fault-tolerant retry logic, and circuit breakers across distributed microservices.
- Built a vector-indexed agent routing registry that eliminated 30% of workflow faults by replacing hardcoded dispatch logic with semantic capability matching — directly improving service quality and availability in a globally distributed Agile team.
- Engineered a CI/CD-triggered RAG ingestion pipeline integrating document chunking, embedding generation, and vector database indexing — automating document processing from upload to production-ready retrieval with zero manual intervention.
- Collaborated with TCS Accessibility Research Team to develop and ship a WCAG-compliant voice agent, applying prompt engineering and NLP techniques to create accessible AI interfaces for users with disabilities.
- Implemented a Human-in-the-Loop (HITL) checkpoint system and an autonomous browser automation agent, both shipped through structured code review cycles and quality gates aligned with responsible AI and security best practices.

BOTSIO Chatbot LLP

Software Engineer Intern – LLM Applications

Mar 2023 – May 2023

- Implemented backend semantic retrieval pipelines — embedding generation, vector indexing, and similarity search — to ground production chatbot responses in verified knowledge sources, improving response consistency for live users.
- Debugged and resolved latency and correctness issues in deployed AI microservices integrated with PostgreSQL; contributed prompt engineering improvements adopted across subsequent model releases.

PROJECTS

Agentic Financial Document Assistant

- Designed a LangGraph-based distributed planner-executor backend for complex query answering over financial documents — planning agent decomposes queries, delegates to specialised sub-agents, and returns schema-validated, cited responses at scale.
- Integrated automated evaluation framework into CI/CD with LLM-as-judge scoring and regression suites, ensuring every deployment meets quality and compliance standards for a regulated BFSI use case.
- Implemented safety guardrails including output schema enforcement, confidence thresholding, and HITL escalation — aligned with responsible AI and enterprise security requirements.

Stack: Python, Django, LangChain, LangGraph, OpenAI API, PostgreSQL, Vector Databases, Docker, REST APIs

LLM Observability & Evaluation Platform

- Built shared backend infrastructure for monitoring LLM services in production: structured logging across prompt versions, token usage, and latency — enabling systematic debugging, cost analysis, and MLOps tracking.
- Reduced evaluation turnaround from hours to minutes by integrating real-time LLM-as-judge scoring and quality alerts directly into CI/CD pipelines.

Stack: Python, LangChain, OpenAI API, PostgreSQL, Docker, CI/CD, REST APIs

RAG-Powered Enterprise Knowledge Assistant

- Deployed horizontally scalable microservices on Azure with hybrid retrieval (dense vector + keyword search) over large document repositories, health checks, autoscaling, and full audit logging for compliance.
- Integrated Azure OpenAI for LLM orchestration with structured output parsing, retry logic, and fallback handling — optimizing performance and cost for enterprise-scale unstructured data processing.

Stack: Python, Django, LangGraph, LlamaIndex, Azure AI Foundry, Azure OpenAI, PostgreSQL, Docker, REST APIs

EDUCATION

Master of Computer Applications (MCA)

2021 – 2023

PES College of Engineering, Mandya

PUBLICATION

AI-Powered Information Retrieval Systems — Peer-reviewed research on semantic search, RAG techniques, and accessible AI interface design. Published 2023.